Lakshya Gupta

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Technical Skills

Languages	C++, Python
GenAI / LLMs	RAG, Hybrid Search (BM25 + Embeddings), Prompt Engineering, Gemini & Groq APIs
AI / ML	Scikit-learn, PyTorch, OpenCV, NumPy, Pandas, LightGBM, XGBoost, Computer Vision
Backend	FastAPI, REST APIs, Pydantic, SQLite, MySQL
Core CS	DSA, OOP, OS, DBMS
Tools	Git, GitHub, Docker, GitHub Actions
Concepts	Testing (pytest), Debugging

Projects

Sensei: RAG-Based JLPT Japanese Tutor
github.com/Lakshya2411/Sensei-JLPT-Tutor

Python · FastAPI · RAG · BM25 · Embeddings · Gemini / Groq · SudachiPy · Docker

- Built a web app that **explains Japanese sentences, generates JLPT practice and corrects learner writing**, grounding LLM answers in 31 hand-written N5–N4 grammar notes.
- Designed **hybrid retrieval** (regex pattern detection + from-scratch **BM25** + **Gemini embeddings**) over grammar notes and mined Tatoeba sentences, fused with **Reciprocal Rank Fusion**; evaluated with Recall@5 and MRR.
- Used **SudachiPy** morphological analysis and **JMdict** for word segmentation, furigana and reading-aware glosses.
- Added a **citation guard** against hallucinated sources, **Gemini** → **Groq** fallback, SQLite caching and per-IP rate limiting; 35 unit and API tests.

AI-Powered Smart Retail & Customer Intelligence Platform
github.com/Lakshya2411/AI-retail

Python · FastAPI · OpenCV · PyTorch · Scikit-learn

- Built an AI-powered retail platform combining **computer vision, sentiment analysis, and a customer support chatbot**.
- Developed face recognition using **Haar Cascades and LBPH** and product classification using **MobileNetV2**.
- Implemented sentiment analysis using **TF-IDF and Logistic Regression** with an ML-assisted FAQ chatbot.
- Created REST APIs using FastAPI and Pydantic, with a Streamlit dashboard, Docker deployment, and GitHub Actions CI/CD.

Fuel Blend Properties
github.com/Lakshya2411/SHELLAI

Python · LightGBM · XGBoost · SciPy · KFold CV

- Blended multi-output and per-target stacks to exploit target correlations and capture per-target nuances.
- Implemented per-target feature selection to select top features per property, reduced MAPE by 5% vs baseline.
- Achieved 2% MAPE reduction through a hybrid ensemble combining multi-output and per-target stacking.

Education

Vellore Institute of Technology — **2023 – 2027**
B.Tech in Computer Science and Engineering — CGPA: 8.16/10.0

Jayshree Periwal High School — **2023**
Science (PCM)

Achievements & Activities

- Cleared Primary Round of Deloitte Hackathon.
- MP Online Course Completed.
- **Languages:** JLPT N4 (Awaiting Certificate) · French (Conversational Fluency)